**LivingEntropyCipher (LEC): A Structure-Independent Cipher with Evolving Entropy
*Full English Version (International Academic Style)***

NS(2026 March)

Abstract

This paper proposes *LivingEntropyCipher (LEC)*, a novel symmetric encryption scheme that does not rely on mathematical hardness assumptions or random number generators. LEC employs a large initial table as its secret key and evolves this table through history-dependent, irreversible mutations during encryption. This "Living Entropy" mechanism enables the cipher to generate near-ideal randomness without using any pseudo-random generator. Experimental results demonstrate that LEC achieves an entropy of 7.99 out of 8.0 and a correlation coefficient of 0.0003 in image encryption, indicating statistical behavior close to true randomness. Due to its structure-independent design and non-reliance on algebraic constructs, LEC is inherently resistant to algebraic, statistical, and quantum-based attacks. This paper details the architecture, mutation model, and security properties of LEC, and evaluates its potential as a new class of volume-based cryptography.

Keywords:LivingEntropyCipher, evolving entropy, structure-independent cipher, irreversible mutation, table-based encryption, quantum-resistant cryptography

1. Introduction

Recent advances in computational power and the emergence of quantum computing have raised concerns regarding the long-term security of cryptographic schemes based on computational hardness. RSA and ECC rely on mathematical structures that may become vulnerable to quantum algorithms such as Shor's algorithm. Even symmetric ciphers like AES and ChaCha20 may be affected by weaknesses in random number generators or structural biases within their internal state transitions.

To address these challenges, this paper introduces *LivingEntropyCipher (LEC)*, a new encryption scheme that does not rely on mathematical structures or random number generators. LEC uses a large initial table as its secret key and applies history-dependent mutations to the table at every encryption step. This design is based on the concept of *Living Entropy*, where entropy evolves dynamically as encryption progresses.

LEC exhibits the following characteristics:

- Structure-independent design with no mathematical hardness assumptions
- No reliance on random number generators
- A large 100 KB table used as the secret key, evolving at every step
- Irreversible, history-dependent mutations
- TableHash-based synchronization and tamper detection
- Achieved entropy of 7.99 (out of 8.0) in experiments
- Achieved correlation coefficient of 0.0003 in image encryption

These properties allow LEC to naturally satisfy Kerckhoffs's principle and provide strong resistance against known attack models, including quantum-based threats. This paper presents the architecture, encryption process, mutation model, and statistical evaluation of LEC, demonstrating its potential as a practical volume-based cryptographic scheme.

2. Background

2.1 Limitations of Computation-Based Cryptography

RSA and ECC rely on mathematical problems such as integer factorization and discrete logarithms. With the development of quantum computing, these problems may be solvable in polynomial time using Shor's algorithm.

2.2 Vulnerabilities of Random Number Generators

Many cryptographic schemes depend on random number generators (RNGs). Biases or leakage of RNG states can lead to critical vulnerabilities.

2.3 Structural Weaknesses of Stream Ciphers

Stream ciphers may suffer from linearity, periodicity, and predictability of internal states, enabling structural attacks.

2.4 Potential of Volume-Based Cryptography

By eliminating mathematical structures and relying instead on large state spaces and history-dependent behavior, volume-based cryptography may offer strong resistance against quantum attacks.

3. Proposed Method: LivingEntropyCipher (LEC)

3.1 Initial Table

LEC uses a 100 KB table as its secret key. This table forms the foundation of the encryption process and evolves continuously at every step.

3.2 Encryption Process

The encryption process consists of the following components:

- cipherBase
- index progression
- XOR operation
- table lookup

These components work together to transform input data into high-entropy output.

3.3 Table Mutation (Evolution)

The core of LEC lies in its history-dependent table mutation.

- Irreversible
- Non-periodic
- History-dependent
- No state is ever repeated

These properties prevent attackers from predicting or reconstructing internal states.

3.4 TableHash

At each step, a hash of the table is computed to ensure synchronization between encryption and decryption.

- Tamper detection
- Man-in-the-middle prevention
- State consistency verification

4. Security Analysis

4.1 Statistical Strength

Image encryption experiments yielded the following results:

- Original image entropy: 5.42 bits/byte
- Encrypted image entropy: 7.99 bits/byte
- Correlation coefficient: 0.0003

These results indicate that LEC produces output statistically close to true randomness.

4.2 Structure Independence

Since LEC does not rely on mathematical structures, algebraic attacks are ineffective.

4.3 No RNG Dependency

LEC does not use random number generators, eliminating RNG-based vulnerabilities.

4.4 Quantum Resistance

LEC has no reversible equations or algebraic structures that quantum algorithms can exploit.

4.5 Kerckhoffs's Principle

The scheme remains secure even when its design is publicly known.

5. Experimental Results

5.1 Image Encryption

- Entropy: 7.99
- Correlation coefficient: 0.0003
- Histogram fully uniform
- Visual output indistinguishable from noise

5.2 Decryption Consistency

Encrypted → decrypted results matched perfectly.

6. Conclusion

This paper proposed *LivingEntropyCipher (LEC)*, a new encryption scheme that does not rely on random number generators or mathematical structures. LEC achieves high entropy and irreversible evolution through a large, history-dependent mutation table. Experimental results show that LEC produces near-random output and exhibits strong resistance to existing attack models. Future work includes formal security proofs, implementation optimization, and exploration of practical applications.